

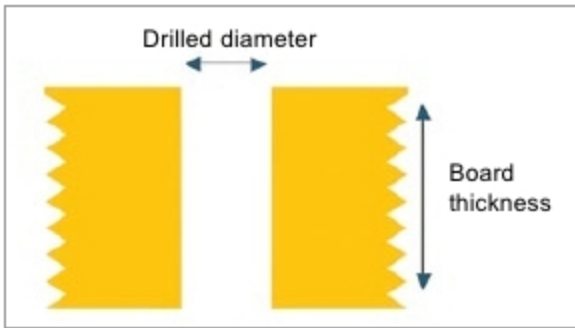


Fig. 5: Aspect ratio of HDI PCB

ing. Glass contained in the pre-pregs changes the laser direction and creates a mediocre or wrong shape quality of laser via holes.

The buildup must use a thinner base copper for good tracks definition. (Many plating processes increase total copper thickness.) The base copper choice also influences the signal propagation for high-frequency applications.

Thinner pre-preg, thinner base copper and thinner PCB are the trends for HDI PCBs.

Via holes are sensitives. Vias are like rivets used during the assembly process. ROHS applies a high thermal stress on vias. Expansion of the material in z-axis stresses the vias. The trend is to continuously reduce via diameter and increase reliability. Further, the assembly process multiplies the number of thermal clocks.

Due to this situation, the only way to reduce the force applied during assembly is to use more stable FR4.

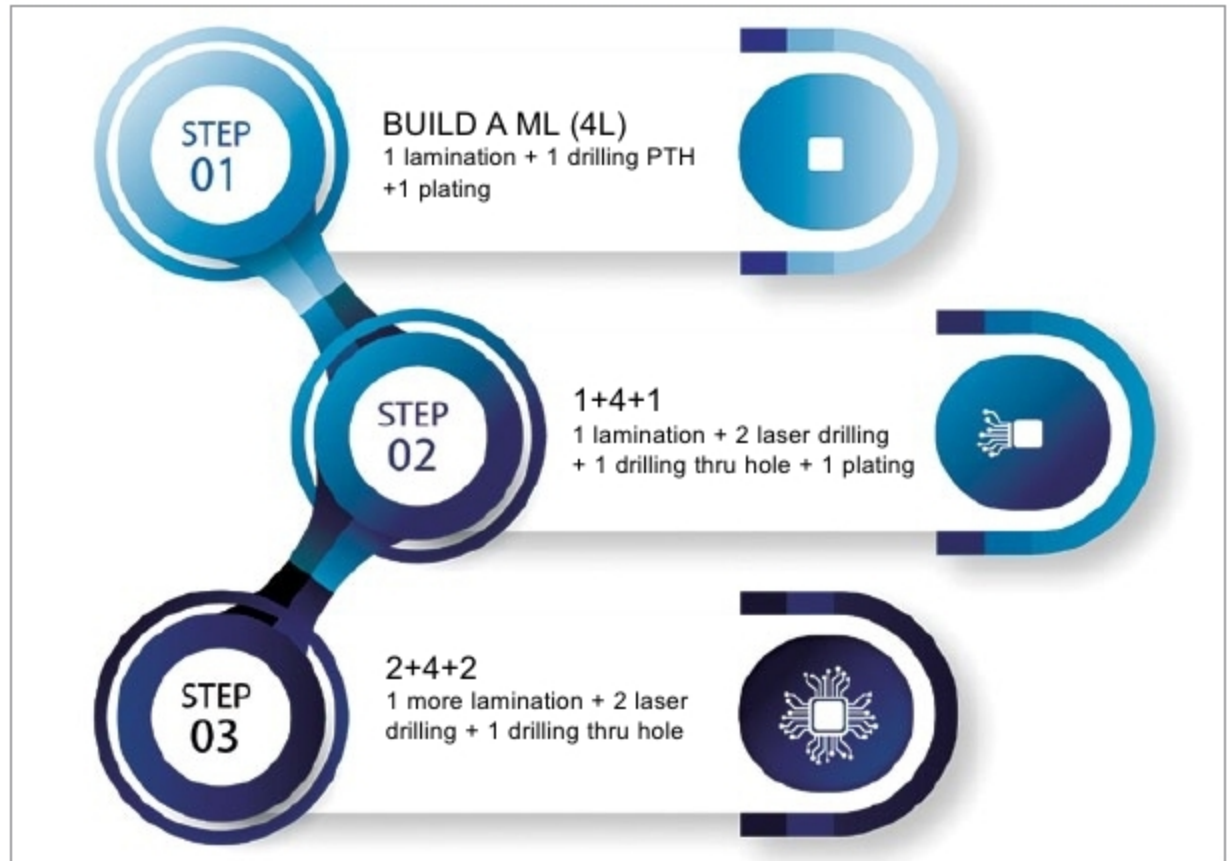


Fig. 6: Lamination process (2+4+2 HDI PCB)

Low CTE (coefficient of thermal expansion) materials are mandatory to limit the breaking holes phenomena during the assembly process and in harsh environments.

FR4 Z CTE, alpha 1 and 2, and Tg are the main parameters to follow for material selection. The real challenge is to select the right material, which is also cost-effective. It is important to reduce the total thickness of boards. Reduction of micro-via diameter pushes towards using thinner pre-pregs and laminates.

Several laser drillings involve a longer lamination process. The lamination process is shown in Fig. 6.

Laser drilling is a sensitive process. The type of chosen pre-preg (web, laser drillable, etc), method of drilling and aspect ratio must be well-studied to get reliable HDI PCB. The R&D department must analyse these new design constraints

carefully. These are detailed below.

CFA phenomena. Right material and layout choice

Via plugging/via filling process. Good design requires clear explanations

Impedance control. DK value, space and lines, dielectric thickness

High-frequency signal propagation in copper. Type of copper versus fine line definition

Advantages of HDI PCB

Advantages of HDI PCBs are given below.

Denser trace routing. It allows incorporation of more components onto smaller boards because HDI PCBs can be populated on both sides of the board.

Enhanced power usage. It decreases power usage, leading to longer battery life in handheld and other battery powered devices.

Smaller weight and size. PCBs fit into little spaces and have a smaller amount of mass than conservative PCB designs. There is lesser chance of harm from mechanical shocks.

Faster signal transmission. A high-density interconnect PCB ultimately results in faster signal transmission along with enhanced signal quality.

Components. It leads to more component space (also by via-in-pad). **EFY**

TABLE II
CAPABILITIES OF HDI PCB

HDI	Capabilities
Number of layers (minimum and maximum)	4-24
HDI builds (1+N+1, 2+N+2)	3+N+3 and any layer
Copper weight (minimum and maximum)	0.5oz~6oz
Minimum track and gap (μm)	65/65μm
PCB thickness (minimum and maximum)	0.4-2.8mm
Dimension (maximum)	457×508mm
Surface finishing (OSEP, ENIG, etc)	HASL, HASL LF, ENIG, immersion tin, OSP, immersion silver, electroplating hard gold/soft gold, gold finger, selective OSP, ENEPIG
Mechanical drill (minimum)	0.15mm
Laser drill (minimum)	0.076mm